

Movie Genre Classification

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Dataset Description & Project Purpose

Understanding data structures

This dataset offers a curated collection of high-quality movie and series poster images from IMDb, each paired with a detailed plot summary (150+ words). It focuses on four main genres: Action, Comedy, Horror, and Romance.

-  **Visual + Textual Data:** Enables **multimodal analysis** by combining posters with rich narrative descriptions.
-  **Genre Classification:** Ideal for developing models that predict genres based on both visual and textual cues.
-  **Applications:** Film studies, machine learning, natural language processing, and recommendation systems.

This dataset serves as a valuable resource for researchers and enthusiasts interested in genre understanding and content-based analysis.

Dataset Description

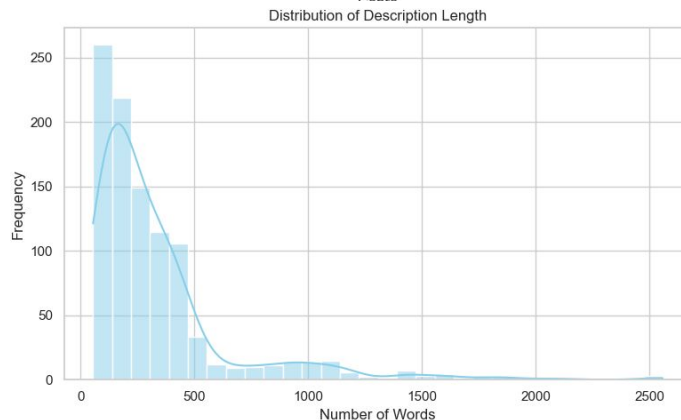
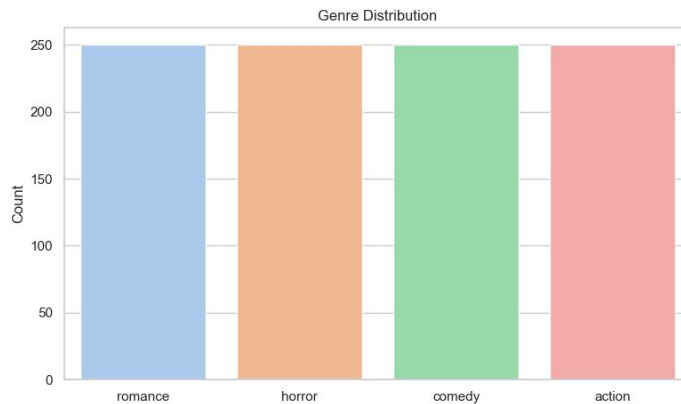
Understanding data structures

Dataset Summary

- **Size:** 1,000 movie/series entries
- **Columns:**
 - **movie_id:** Unique ID
 - **description:** Original plot summary
 - **genre:** One of four genres (Action, Comedy, Horror, Romance)

Genre Distribution

- Romance: 250
- Horror: 250
- Comedy: 250
- Action: 250



Text Preprocessing Pipeline

Preprocessing Description Column

To prepare plot summaries for modeling, we applied the following NLP cleaning steps:

1. **Lowercasing** – Convert all text to lowercase
2. **Remove Noise** – Eliminate punctuation and extra whitespace
3. **Tokenization** – Break text into individual words
4. **Stopword Removal** – Filter out common English words (e.g., "the", "is", "and")
5. **POS Tagging + Lemmatization** –
Normalize words based on their part of speech (e.g., *running* → *run*)
6. **De-duplication** – Remove consecutive duplicate tokens

Output stored in new column: `clean_description`

This cleaned text is used for feature extraction and modeling.



LSA-Based Text Summarization Tool

Preprocessing Description Column

We developed a lightweight **text summarization tool** based on **Latent Semantic Analysis (LSA)** using the following approach:

Method Overview:

- **Sentence Tokenization**

The input plot summary is split into individual sentences.

- **TF-IDF Vectorization**

Each sentence is vectorized using the **Term Frequency–Inverse Document Frequency** (TF-IDF) method, which emphasizes important and distinctive words.

- **Dimensionality Reduction via SVD**

We apply **Truncated Singular Value Decomposition (SVD)** with one component (`n_components=1`) to capture the main semantic topic across all sentences.

- **Salience Scoring & Selection**

Sentences are ranked based on their contribution to the main semantic topic. The top `k` sentences (default: 2) are selected to form the summary.

- **Summary Output**

Selected sentences are ordered based on their original position to maintain coherence and returned as the final summary.

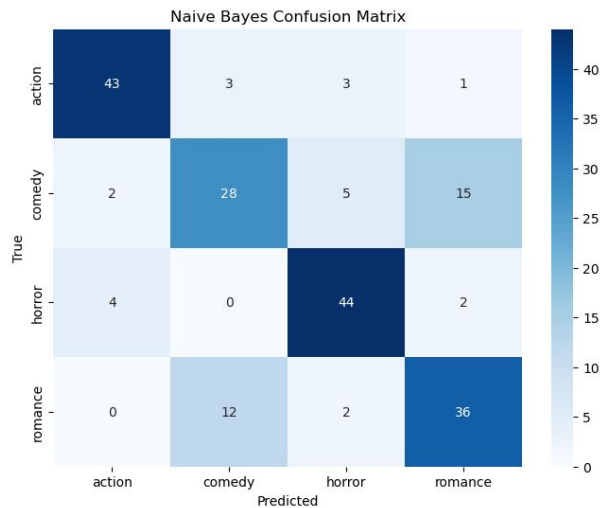
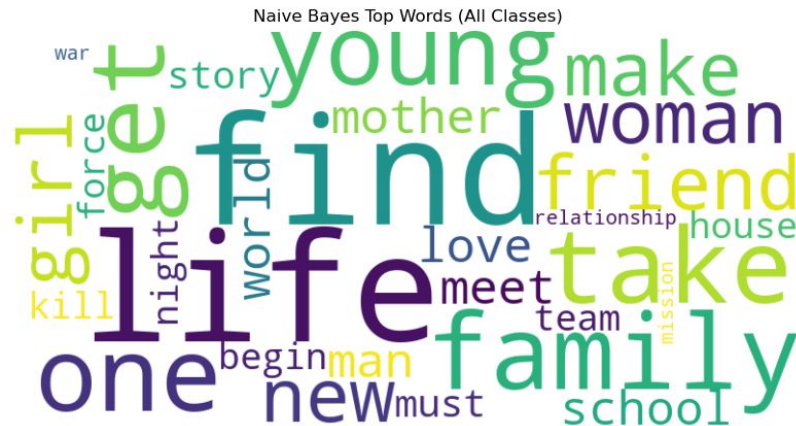
Naive Bayes (Text-based)

Model Summary

- Input: Cleaned plot summary (`clean_description`)
- Feature Extraction: TF-IDF (`max_features = 5000`)
- Best performance:
Macro F1 = 0.7523 (best among all tested settings)

Key Insights

- Simple yet effective for text classification
- Strong in genre-specific vocabulary identification
- Best performance when trained on the cleaned plot description only



Random Forest (Text-based)

Model Summary

- Input: Combined `summary` + `clean_description`
- Feature Extraction: TF-IDF (`max_features` = 4000)
- Best performance:
Macro F1 = 0.6937

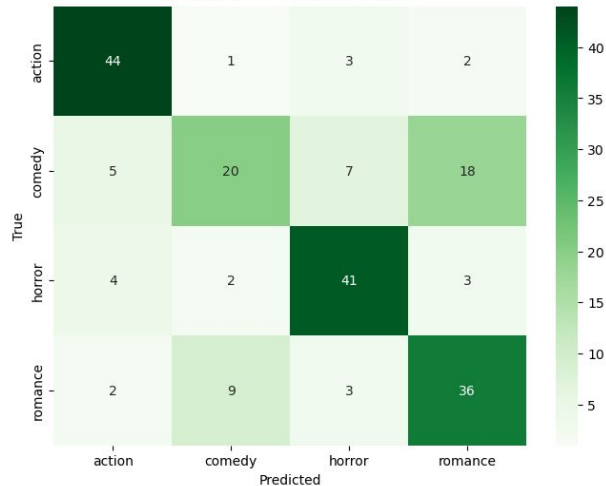
Key Insights

- Captures non-linear relationships
- Performs well when both raw and cleaned summaries are combined
- Slightly lower F1 than Naive Bayes, but more robust to noise

Random Forest Important Words (Top 20)



Random Forest Confusion Matrix

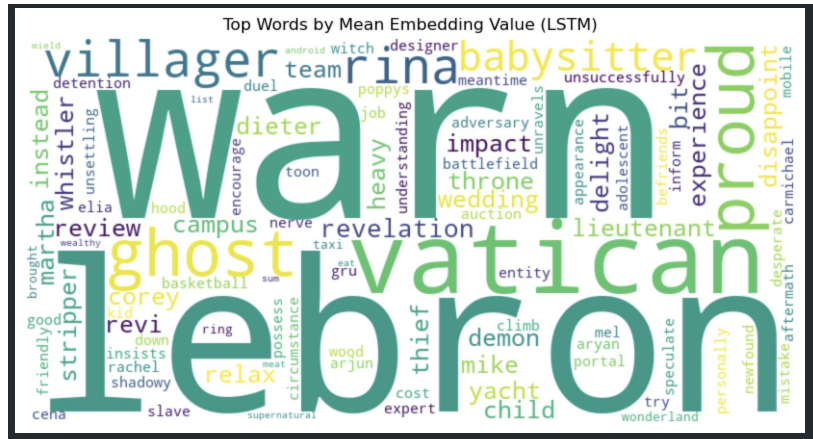


LSTM (Text-based)

Model Summary

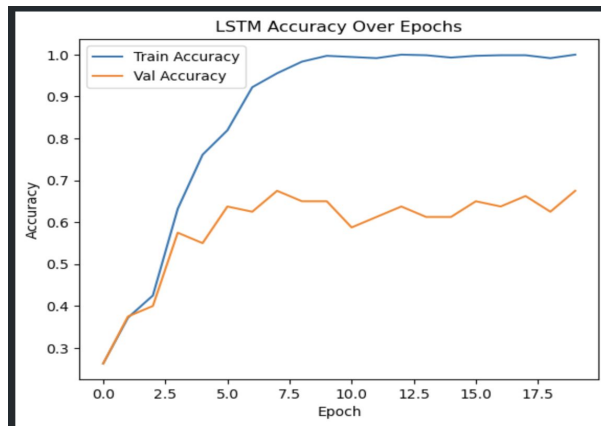
- Input: Cleaned summary (`clean_description`)
- **Embedding:** Tokenizer + padded sequences (maxlen = 300)
- **Model structure:**
- `Embedding(5000, 64) → LSTM(64) → Dropout(0.5) → Dense(32, relu) → Dense(4, softmax)`
- **Best performance:**

Macro F1 = 0.58 ; Accuracy = 0.575



Key Insights

- LSTM captures temporal patterns and sequential dependencies in the text
- Higher recall in Romance, high precision in Horror — model shows genre-specific bias
- Word embeddings enable semantic understanding, but model struggles with noisy or short input



User Interface (UI)

- Input a movie description
- Frontend (Streamlit)
 - a. Take in raw movie description
 - b. Process to get summary and cleaned text
 - c. Send HTTP Post request to backend to get results
 - d. Display results from 4 ML models and use majority vote to get final prediction
- Backend (FastAPI)
 - a. Load in pre-trained models and vectorizers/tokenizer
 - b. Transform input text and predict class + probability
 - c. Wrap up results in a dictionary

Movie Genre Classification: Compare 4 ML Models

Enter a movie description to get genre predictions from 4 different models.

📄 Movie Description

Four years after the catastrophic events that led to the destruction of Isla Nublar, the world is now a drastically changed place. Dinosaurs, once confined to the isolated ecosystems of the island, are no longer simply a distant memory or an attraction in a theme park—they are part of the global landscape. These ancient creatures roam free, inhabiting diverse environments and hunting alongside humans in an uneasy coexistence that constantly teeters on the edge of chaos.

As dinosaurs find their place in a rapidly evolving ecosystem, the delicate balance between mankind and prehistoric predators becomes more fragile. The human race must confront its place on a planet that no longer belongs solely to them. With species both old and new adapting to the modern world,

 Classify

Predictions:

🔍 Naive Bayes
Predicted: **action**
Probability: 31.50%

🔍 Random Forest
Predicted: **action**
Probability: 40.00%

🔍 Logistic Regression
Predicted: **action**
Probability: 35.14%

🔍 Lstm
Predicted: **action**
Probability: 36.72%

✅ Final Prediction (Majority Vote):
action

Bonus: Poster-Based Genre Classification

Workflow & Modeling Approach

Data Preprocessing:

Resized all posters to 224×224

Normalized pixel values

Split into train (80%) and validation (20%)

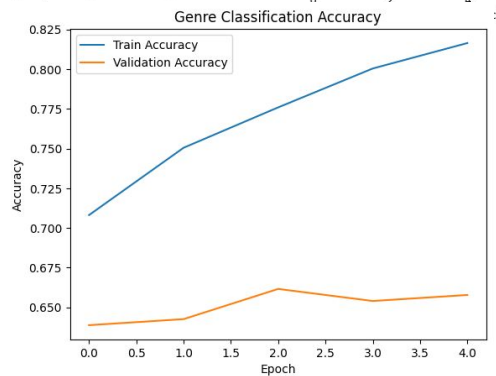
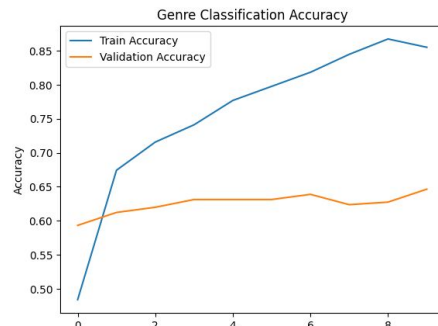
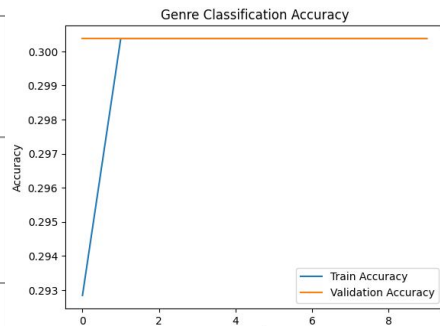
Modeling Pipeline:

Model 1 (Base): MobileNetV2 with random weights

Model 2 (Pretrained): MobileNetV2 with frozen ImageNet weights

Model 3 (Fine-Tuned): Top 50 layers unfrozen for additional learning

Result:	Train Acc	Val Acc	Observation:
Base Model	~32%	~30%	Failed to learn — trained from scratch
Pre-trained	~84%	~64%	Good generalization with frozen layers
Fine-Tuned	~86%	~66%	Improved slightly via deep fine-tuning



Bonus: Results, Interpretation & Next Steps

Performance of the Fine-Tuned Poster Classification Model

Genre	Precision	Recall	F1-Score	
Action	0.63	0.63	0.63	Overall accuracy: ~66%
Comedy	0.68	0.77	0.72	
Horror	0.70	0.77	0.73	
Romance	0.57	0.40	0.47	

Best genres: Comedy and Horror

Weakest genre: Romance (low recall)

Model Interpretation:

- Pretrained MobileNetV2 provided strong baseline features
- Fine-tuning improved genre-specific recall, especially for Comedy & Horror
- Romance misclassifications likely due to:
 - Visual inconsistency
 - Fewer clear thematic patterns in posters

Future improvements:

Multimodal Learning: Combine posters with plot summaries to leverage both visual and textual cues for better genre understanding.

Class Balancing: Apply class weights or oversample underrepresented genres (e.g., Romance) to reduce bias and improve recall.

Data Augmentation: Enhance generalization by diversifying poster appearances through techniques like rotation, zoom, and color jitter.

THANK YOU!

Questions or Feedback?